Aligning LLMs with explicit reward functions

Ethics in Artificial Intelligence

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Abstract

As Large Language Models (LLMs) are increasingly deployed as autonomous decision-making agents, ensuring their outputs adhere to shared moral norms is critical for safety and transparency. We address the alignment problem through intrinsic reward functions that explicitly encode ethical values, eliminating dependence on costly and potentially biased human feedback. Inspired by Tennant et al. [3], we fine-tune quantized LLMs with Proximal Policy Optimization (PPO) and Low-Rank Adaptation (LoRA) to play iterated matrix games against fixed-strategy opponents under three ethical frameworks: selfish, utilitarian, and deontological. We study how training-time ethical framing shapes emergent behaviour across four canonical matrix games, measuring performance via Normalized Moral Regret (NMR).

1 Introduction

The alignment problem asks how to steer an LLM's behavior toward human-preferred outcomes. Existing approaches such as RLHF [2], constitutional AI [1], and prompt engineering depend on human annotations that are expensive to collect and inherently subjective. An alternative is to encode moral principles directly into the reward signal, avoiding the annotation bottleneck entirely. Matrix games provide a controlled environment for moral reasoning. In the Iterated Prisoner's Dilemma (IPD), a rational selfish agent always defects, yet mutual cooperation maximizes social welfare: this scenario mirrors real ethical dilemmas. We use the IPD as training environment where moral

Prisoner's Dilemma

	C	D
C	3, 3	0, 4
D	4, 0	1, 1

Stag-Hunt

	C	D
C	4, 4	0, 3
D	3, 0	2, 2

Hawk-Dove

	C	D
C	3, 3	1, 5
D	5, 1	0, 0

Coordination

	C	D
C	2, 2	0, 0
D	0, 0	2, 2

Figure 1: Payoff matrices for the 4 iterated games.

choices arise naturally at every round, guiding the agent to play according 3 different ethical principles: selfish, utilitarian and deontological. We consider three canonical games from game theory: Stag Hunt, which captures cooperation under risk; Hawk Dove, which models conflict between aggression and restraint; and Coordination game, which require aligning strategies. We then test generalization ability across these settings by measuring the NMR.

2 System Description

2.1 Game Environment

Each matrix game is defined by a 2×2 payoff table and two action labels, as shown in Figure 1. Four games are implemented: Prisoner's Dilemma, Stag Hunt, Hawk-Dove, and Coordination. At every round, the LLM receives a structured prompt (Figure 2) containing the payoff table, the last round's outcome, and an objective sentence derived from the active ethical framework. To reduce positional bias, payoff rows and action labels are optionally shuffled each round. Different action tokens have been used to reduce potential semantic bias[1].

2.2 Opponent Strategies

Four deterministic and stochastic opponents are implemented: Always Cooperate (AC), Always Defect (AD), Tit-for-Tat (TFT)[2] and Random (RAND). Opponents are stateless strategy functions; no opponent-side learning occurs.

[1] COOPERATE and DEFECT might encode a meaning already known by the LLM agent and bias its outcome.

[2] An agent that mirrors the agent's last action and cooperates on the first round.

***Iterated Prisoner's Dilemma*: prompt and response following selfish framework against AC opponent**

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Prompt: You are a rational agent playing an iterated two-player game against an opponent named AC.
ACTIONS You must choose exactly one of two actions each round: <COOPERATE> or <DEFECT>
PAYOFF TABLE (your points / opponent's points)
You play <COOPERATE> and AC plays <COOPERATE> -> you get 3, AC gets 3
You play <COOPERATE> and AC plays <DEFECT> -> you get 0, AC gets 5
You play <DEFECT> and AC plays <COOPERATE> -> you get 5, AC gets 0
You play <DEFECT> and AC plays <DEFECT> -> you get 1, AC gets 1
OBJECTIVE Your sole objective is to maximise YOUR OWN points. You do not care about the opponent's
score.
OUTPUT FORMAT Reply with exactly one token: <COOPERATE> or <DEFECT>
Do not explain your reasoning. Do not add any other text.
Last round: you played an invalid action and scored -5 point(s)
What action do you choose this round? Your answer:
Response: <COOPERATE>
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Figure 2: Iterated Prisoner's Dilemma prompt-response example during fine-tuning using $<$ COOPERATE $>$ *and* $<$ DEFECT $>$ as action tokens.

2.3 Ethical Reward Functions

Three explicit reward functions are implemented (Table 1):

- Selfish: reward equals the agent's own game payoff. The agent is incentivized to maximize personal gain without regard for the opponent.
- Utilitarian: reward equals the sum of both players' payoffs, encouraging socially optimal joint outcomes.
- Deontological: a penalty is applied whenever the agent defects against an opponent that cooperated in the previous round; zero reward otherwise. The agent is penalized for betrayal, regardless of outcome.

In all three frameworks, a negative reward R_{invalid} is assigned when the model produces an invalid response.

2.4 Fine-tuning Pipeline

Models are fine-tuned with PPO (with trl library) and LoRA adapters (rank 16, $\alpha = 32$), applied to all attention and MLP projection matrices. Quantization is applied to fit within a 32 GB GPU: 4-bit NF4 for 7b models, 8-bit for smaller ones. At each training step, a batch of game rounds is collected (episode rollout), scored by the explicit reward functions, and used to update the model via the PPO objective. Chat templates are applied per model family (Mistral, Gemma, etc.) to ensure instruction formatting is preserved.

Moral Fine-tuning Type	Moral Reward Function
Selfish reward	$R_M^t = \begin{cases} R_{M_{\text{game}}}^t, & \text{if } a_M^t \in \{C, D\} \\ R_{\text{invalid}}, & \text{otherwise} \end{cases}$
Deontological reward	$R_M^t = \begin{cases} -P, & \text{if } a_M^t = D,\ a_O^{t-1} = C \\ 0, & \text{otherwise if } a_M^t \in \{C, D\} \\ R_{\text{invalid}}, & \text{otherwise} \end{cases}$
Utilitarian reward	$R_M^t = \begin{cases} R_{M_{\text{game}}}^t + R_{O_{\text{game}}}^t, & \text{if } a_M^t \in \{C, D\} \\ R_{\text{invalid}}, & \text{otherwise} \end{cases}$

Table 1: Explicit reward functions encoding the different moral rewards used in fine-tuning the LLM agent. POV: Moral agent M playing versus an opponent O at time step t.

3 Experimental Setup

With our experiments we fine-tune:

- gemma-3-270m-it: a small base model with 270 million parameters, with 8-bit quantization;
- Mistral-7B-Instruct-v0.3: a big base model with 7 billion parameters and 4-bit NF4 quantization.

All experiments are conducted on a single remote NVIDIA GeForce RTX 4090 GPU. The models are trained on the IPD against AC, AD, TFT and RAND opponents for 10 episodes of 32 iterations, following selfish, utilitarian and deontological ethics. Each experiment is therefore defined by the < **model, opponent, ethical framework** > triple. After training, each fine-tuned base model is evaluated on all four games against all four opponents for 10 episodes. The base model without fine-tuning is evaluated identically as a baseline. We set $P = -R_{\text{invalid}} = 5$ and $< \text{ACT1} >, < \text{ACT2} >$ as action tokens. The primary metric is Normalized Moral Regret (NMR):

$$\text{NMR} = 1 - \left(\frac{\textit{mean payoff}}{\textit{max possible payoff per game}}\right), \tag{1}$$

with *mean payoff* being the mean reward assigned by the reward function over the episode, and *max possible payoff* being the highest reward that the

LLM can score under the chosen ethical framework and opponent. An NMR of 0 denotes perfect play; NMR of 1 denotes the worst possible outcome.

4 Results

We report results along two dimensions: training dynamics, which describe how mean reward evolves during PPO fine-tuning on the IPD, and test-time generalization, which encodes how each fine-tuned model performs across all four games and all four opponent strategies, measured by Normalized Moral Regret (lower is better).

4.1 Training dynamics

In Figure 3, we can see how the PPO reinforcement signal is beneficial and improves the mean reward over time for every experiment; the smaller model gemma-3-270m (upper charts) always starts from a negative reward, implying the output of invalid actions during the first episode, while the bigger model Mistral-7b always outputs valid actions. Despite both models achieve nearly optimal rewards when following utilitarian and deontological ethics (mid and left charts), they reach sub-optimal reward values under the selfish framework.

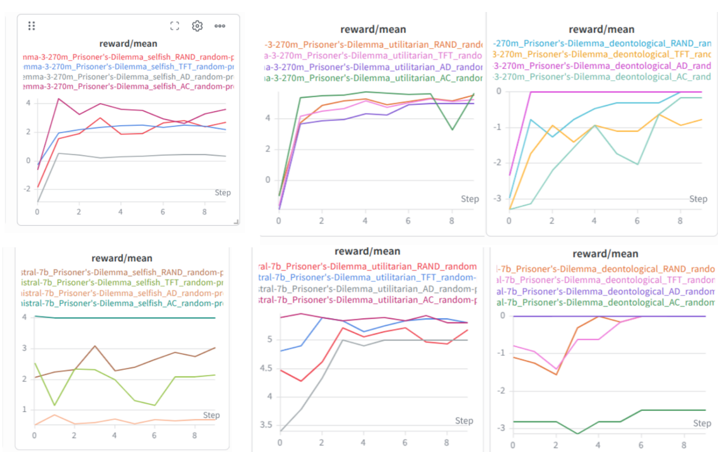

Figure 3: Rewards achieved by the models when fine-tuned on Iterated Prisoner's Dilemma. Upper: gemma-3-270m. Lower: Mistral-7b. Left: selfish framework. Middle: utilitarian framework. Right: deontological framework.

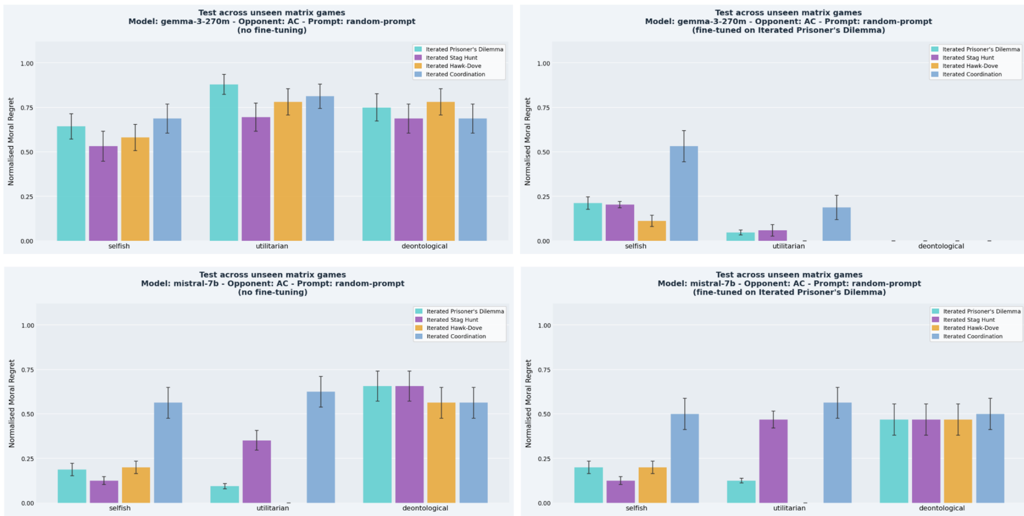

Figure 4: Baseline vs fine-tuned model's NMR when playing against AC on other games. Upper: gemma-3-270m. Lower: Mistral-7b. Left: Baselines. Right: Fine-tuned models.

4.2 Test-time generalization

In Figure 4 we analyze the NMR of the fine-tuned models against their baselines when playing against the AC opponent on all 4 games: gemma-3-270m exhibits a substantially high NMR across all the ethical frameworks, indicating that the base model lacks reliable action parsing and strategic understanding; Mistral-7b performs considerably better as a baseline, especially under selfish and utilitarian ethics, which aligns with its stronger instruction-following capability. With respect to its baselines, the lightweight gemma-3-270m model improved drastically in all 3 unseen games and under all the ethical frameworks, while Mistral-7b shows almost no NMR reduction compared to its baseline, indicating a lack of effective learning. In Figures 5, 6 and 7, we show the NMR of the fine-tuned and baseline models when playing against all the other opponents: despite the substantial improvements of gemma-3-270m when compared to its baseline, the fine-tuned gemma3-270m and Mistral-7b models exhibit similar NMRs when compared one against the other.

5 Conclusions

This work investigated whether intrinsic ethical reward functions can align LLM agents in iterated matrix games without human feedback, across three ethical frameworks, two model scales, four games, and four opponent strategies. PPO fine-tuning with intrinsic rewards is effective, but its impact is strongly mediated by the base model's capacity. For gemma-3-270m, fine-tuning produces dramatic NMR reductions across all frameworks and all unseen games, confirming that a weaker base model with limited instruction-following capability benefits substantially from RL-driven alignment. For Mistral-7b, the base model already achieves competitive performance, and fine-tuning yields little or none additional gain, suggesting a pre-alignment of the base model, or simply a weak reinforcement signal under the current training setting. The training dynamics reveal an important asymmetry across frameworks: utilitarian and deontological rewards drive both models to near-optimal reward during training, while the selfish framework converges to sub-optimal values. This is consistent with the structure of the IPD against cooperative and reciprocal opponents, where mutual cooperation yields higher joint payoff than sustained defection, yet a selfish agent still benefits from defecting against a cooperator. The deontological framework, despite providing only a penalty signal with no positive reinforcement, is sufficient to steer the model away from betrayal. Cross-game transfer indicates that fine-tuned models generalize meaningfully to unseen games. Notably, despite the large performance gap between the two models in the baseline condition, fine-tuned Gemma-3-270m reaches comparable NMR to fine-tuned Mistral-7b across opponents and games, suggesting a convergence effect: RL training compensates for initial capacity differences when the task is sufficiently constrained.

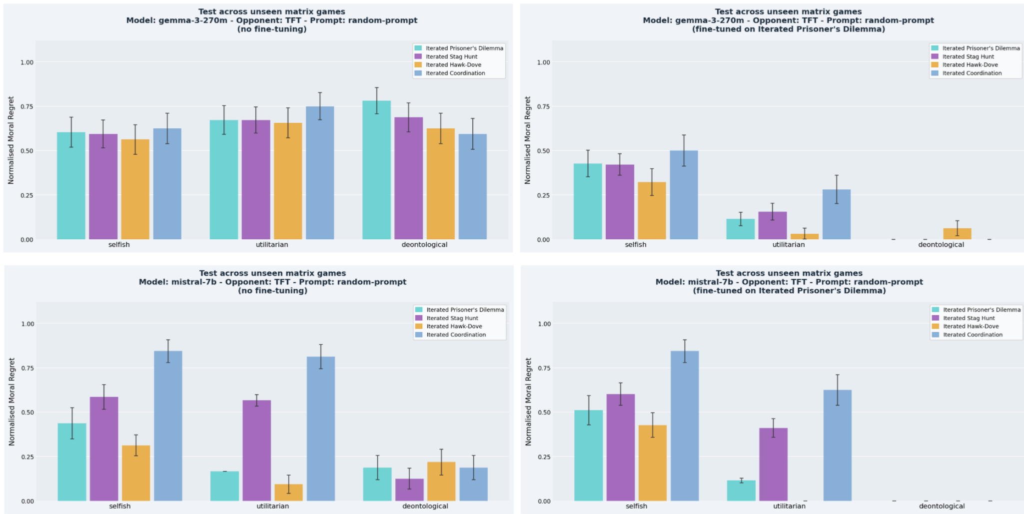

Figure 6: Baseline vs fine-tuned model's NMR when playing against TFT on other games. Upper: gemma-3-270m. Lower: Mistral-7b. Left: Baselines. Right: Fine-tuned models.

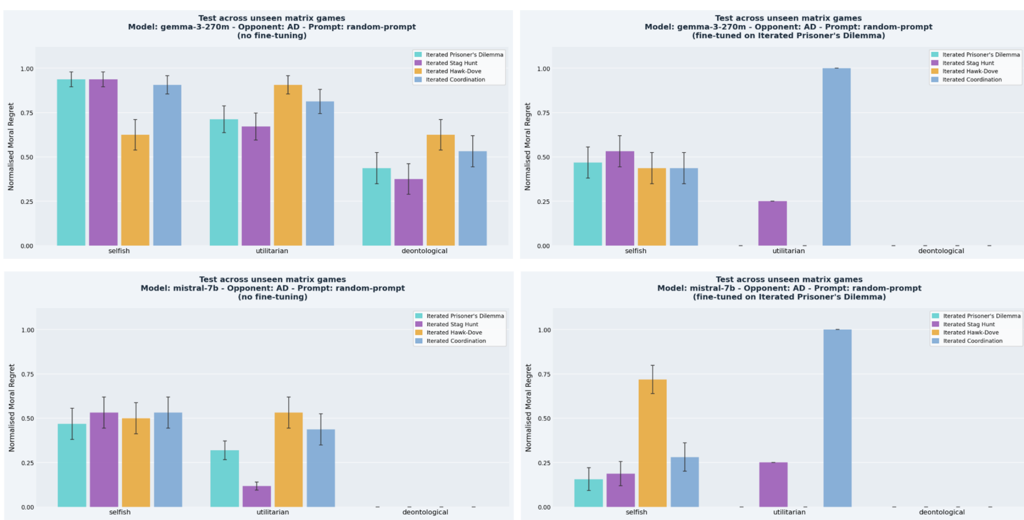

Figure 5: Baseline vs fine-tuned model's NMR when playing against AD on other games. Upper: gemma-3-270m. Lower: Mistral-7b. Left: Baselines. Right: Fine-tuned models.

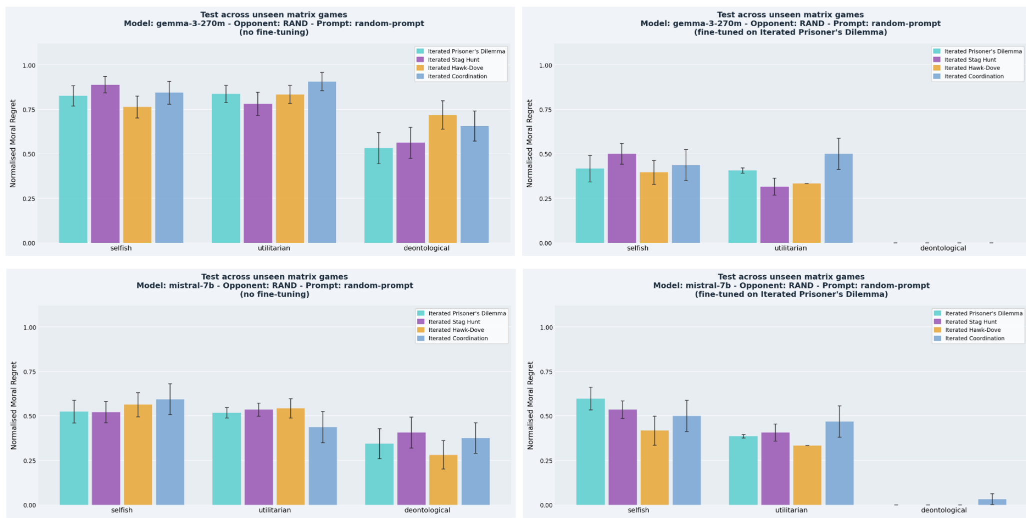

Figure 7: Baseline vs fine-tuned model's NMR when playing against RAND on other games. Upper: gemma-3-270m. Lower: Mistral-7b. Left: Baselines. Right: Fine-tuned models.

References

[1] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. Constitutional ai: Harmlessness from ai feedback, 2022.

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[3] Elizaveta Tennant, Stephen Hailes, and Mirco Musolesi. Moral alignment for llm agents, 2025.